

MACHINE LEARNING ON GRAPHS

UE23CS343BB4

Mini-Project

Graph-Based Location Recommendation System using Graph Attention Networks (GAT)

TEAM:

HARISH SAJJAN	PES1UG23CS230
GOUTHAM N	PES1UG23CS220
KUSHAL D	PES1UG23CS318

1. Problem Statement

In modern recommendation systems, traditional approaches often fail to capture the complex relationships between users and items. This project addresses the problem of recommending relevant locations (such as waterfalls, temples, beaches, etc.) to users based on their past interactions.

The objective is to design a system that models users and locations as a graph and leverages Graph Neural Networks (GNNs), specifically Graph Attention Networks (GAT), to generate personalized and accurate recommendations.

2. Relevance of the Problem

Recommendation systems are widely used in:

- travel platforms (Google Maps, TripAdvisor)
- e-commerce (Amazon)
- content platforms (Netflix, YouTube)

However, traditional systems:

- ignore relationships between entities
- fail to capture structural dependencies

Graph-based approaches:

- model real-world relationships naturally

- improve recommendation accuracy
- support personalization

This makes the problem highly relevant in real-world applications.

3. Literature Review

3.1 Traditional recommendation systems include:

Content-Based Filtering

- Uses item features
- Limitation: cannot capture user-user or item-item relationships

Collaborative Filtering

- Uses user-item interactions
- Limitation: suffers from sparsity and cold-start problems

Graph-Based Methods

Recent research shows:

- Graph Neural Networks (GNNs) improve recommendations
- Models like:
 - Graph Convolutional Networks (GCN)
 - Graph Attention Networks (GAT)

Graph Attention Networks (GAT)

- Introduced attention mechanism
- Assigns importance to neighbours
- Handles complex relationships better

This project uses GAT to improve recommendation quality.

4. Dataset Collection & Preprocessing

4.1 Dataset

A synthetic dataset is generated consisting of:

- 5000 places
- 200 users

4.2 Place Features

Each place contains:

- type (waterfall, temple, etc.)
- latitude
- longitude
- rating

4.3 User Data

- Users are connected to places they visited
- Each user visits 10 places

4.4 Preprocessing Steps

1. Encode place types using label encoding
2. Normalize features (implicit through model)
3. Construct graph edges:
 - user → place interactions
 - place → place (nearest neighbors)

5. Proposed Solution

5.1 The system builds a graph:

Nodes:

- Users
- Places

Edges:

- User → Place (visited)
- Place → Place (similar/nearby)

5.2 Model: Graph Attention Network (GAT)

Architecture:

- Input: 4 features
- Layer 1: GATConv (64 features, 2 heads → 128)
Head 1 → focuses on spatial similarity
Head 2 → focuses on type similarity
- Activation: ReLU
- Layer 2: GATConv (128 → 64)

5.3 Working

1. GAT learns node embeddings using graph structure
2. Attention mechanism assigns importance to neighbors
3. Each node gets a meaningful representation

5.4 Recommendation Logic

Step 1: Compute similarity

- cosine similarity between user and place embeddings

Step 2: Apply hybrid filtering

- frequency-based weighted preference

$$FinalScore = Similarity + (0.1 \times TypeWeight)$$

Step 3: Ranking

- sort places by final score
- select top-K places

5.5 Explainability

System provides reasons:

- similar type
- nearby location
- graph similarity

5.6 Visualization

- Map visualization using Folium
- Graph visualization using NetworkX

6. Results and Experimentation

Training Results

Epoch 0 Loss: 119

Epoch 10 Loss: 8.45

Epoch 20 Loss: 1.72

Observations

- GAT successfully captures user preferences
- Recommendations reflect:
 - user behavior
 - graph structure

Key Findings

- Frequency-based preference improves personalization
- GAT captures deeper relationships beyond type
- Hybrid approach balances accuracy and diversity

7. Conclusion

This project demonstrates the effectiveness of Graph Attention Networks in recommendation systems. By modeling users and places as a graph, the system captures both structural and behavioral relationships.

The integration of:

- GAT embeddings
- cosine similarity
- hybrid preference weighting

results in a robust and scalable recommendation system.

Future Work

- Use real-world datasets
- Add user features (age, preferences)
- Improve diversity control
- Deploy as web application